

SADEV ABEYKOON

Undergraduate | BICT(Hons) - Rajarata University of Sri Lanka
Targeting: IoT & Embedded Systems Internship

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PROFESSIONAL PROFILE

Final-year BICT(Hons) undergraduate specializing in IoT and Embedded Systems, with hands-on experience in ESP32, Arduino, and Raspberry Pi hardware, and firmware development across I2C, I2S, and MQTT protocols. Currently conducting active research in multimodal edge AI fault diagnosis, bridging sensor integration with TensorFlow Lite deployment on edge hardware. Seeking an IoT or Embedded Systems internship to contribute firmware and hardware integration skills in a production environment.

EDUCATION

BICT (Hons) in Information Communication Technology - Current GPA: 3.48/4.0

Rajarata University of Sri Lanka - 2022 – Present (Following)

- **Core modules:** Embedded Systems, Internet of Things, Robotics, and Artificial Intelligence.

G.C.E. Advanced Level – 2020

Successfully completed the GCE Advanced Level examination in the Engineering Technology stream, achieving 1B and 2Cs.

WORK EXPERIENCE

Associate Image Processor

OREL IT (PVT) Ltd · Colombo, Sri Lanka - February 2021 – October 2022

- Annotated and labelled images for AI services using CVAT (Computer Vision Annotation Tool), supporting object detection and image classification model training.
- Processed high volumes of image data with strong attention to detail, maintaining consistent accuracy across repetitive annotation tasks.
- Contributed to building quality training datasets that directly feed machine learning model development pipelines

RESEARCH

Multimodal Edge AI for Automotive Engine Fault Diagnosis

Ongoing Research | Rajarata University of Sri Lanka - December 2025 – Present

- Developing a low-cost portable AI diagnostic system using the Toyota 1NZ-FE engine as a case study, fusing synchronized Audio (INMP441) and Vibration (MPU-6050) sensor data to distinguish between Tappet Noise, Timing Chain Elongation, and Misfires
- Designing and evaluating a Dual-Stream CNN model for multimodal sensor fusion, to be deployed on Raspberry Pi using TensorFlow Lite (INT8 quantization) for real-time edge inference without cloud dependency
- Technologies: Raspberry Pi 4, INMP441, MPU-6050, Python, TensorFlow/Keras, TensorFlow Lite, CNN.

UNIVERSITY PROJECTS

Smart Parking System – 2025

- IoT-based system with ESP32, IR sensors, Firebase real-time sync, QR-based gate control, online booking, and cashless payment; achieved 100% pass rate across 45 test cases.
- Technologies: ESP32, Arduino IDE, PHP, MySQL, Firebase, HTML, CSS, JavaScript.

AutoClean – Autonomous Cleaning Robot - 2025

- Autonomous floor cleaning robot with ultrasonic obstacle avoidance, DC suction, and Wi-Fi remote control via mobile app.
- Technologies: Arduino Uno, ESP32, L298N motor driver, Arduino IDE.

AI Voice Assistant - 2024

EXTRU 2024 Exhibition – Rajarata University- **GitHub:** github.com/Sadev-Hiruditha/voice_Assistant.

- Built a Python-based AI voice assistant using spaCy for NLP intent detection, supporting Wikipedia search, YouTube playback, Google search, and application control via a Tkinter GUI
- Technologies: Python, spaCy, SpeechRecognition, pyttsx3, pywhatkit, Tkinter.

TECHNICAL SKILLS

- Microcontrollers & Hardware: ESP32, Arduino Uno, Raspberry Pi 4, INMP441, MPU-6050, IR Sensor, Ultrasonic HC-SR04
- Protocols & Firmware: I2C, I2S, UART, SPI, Wi-Fi, MQTT, Arduino IDE, C/C++, Python
- Edge AI & Tools: TensorFlow Lite (INT8), TensorFlow/Keras, CNN, FFT/STFT, Git, GitHub, Firebase, MySQL
- Languages: English, Sinhala, German (Basic).

EXTRA-CURRICULAR ACTIVITIES & LEADERSHIP

- Project Mentor – EXTRU 2025 Exhibition (under ATIT, Faculty of Technology, RUSL): Mentored two student project groups: Node.js-based AI Voice Assistant with voice-controlled robotic arm and AI-powered Checkers game using the Minimax algorithm
- Member – ATIT Rajarata, Faculty of Technology, Rajarata University of Sri Lanka
- Editor – PEDONE Chess Society, Rajarata University of Sri Lanka: Responsible for society communications, publications, and media content
- Faculty Chess Team member since 2022 and contributed to the team's victory at the 2024 Men's Chess Championship at the Inter-Faculty Meet.

REFEREES

Prof. Prasad Tharanga Jayasooriya

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Technology, Faculty of Technology, Rajarata
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